

BACKGROUND

Clone fitness has so far been measured by **watching** a tumour evolve: repeated single-cell sequencing of cell lines and xenografts, then time-series modelling (fitClone, Salehi et al. 2021). A patient, by contrast, is usually sequenced once: a **single snapshot** of the tumour, not a time series.

Question

From one clonal copy-number snapshot, can we recover the selection coefficient of every chromosome arm, $\theta \in \mathbb{R}^{44}$, with its uncertainty? There is **no likelihood** — tumour evolution exists only as a simulator — and **extinct clones are unseen**.

Why arm-level θ ?

A cell's **fitness** — its odds of dividing rather than dying — is set by its copy numbers:

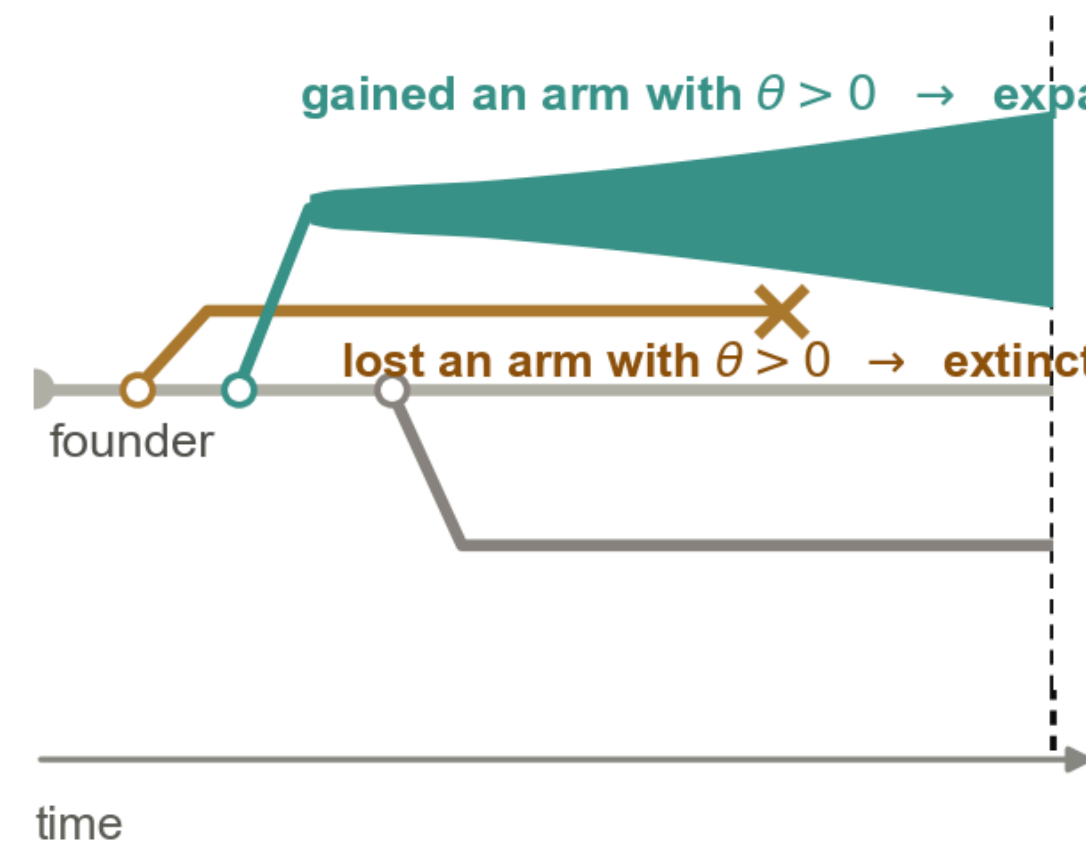
$$\text{fitness} = \prod_{\text{arms}} (1 + \theta_{\text{arm}})^{\text{copies} / \text{ploidy}}$$

θ_{arm} is the weight of one extra copy of that arm: $\theta > 0$ favours gains, $\theta < 0$ favours losses. Fitness is a property of *one clone*; θ is a property of the *arm*, so it

- **scores any clone** from its copy numbers — including clones not yet seen;
- **predicts the future** of the tumour: which clone takes over, what a therapy leaves behind;
- **ranks the arms** that drive this cancer — the fitness landscape, comparable across patients;
- **comes with uncertainty**: the posterior says which arms are established and which are guesses.

What actually happened, what one biopsy shows, and what we want to know

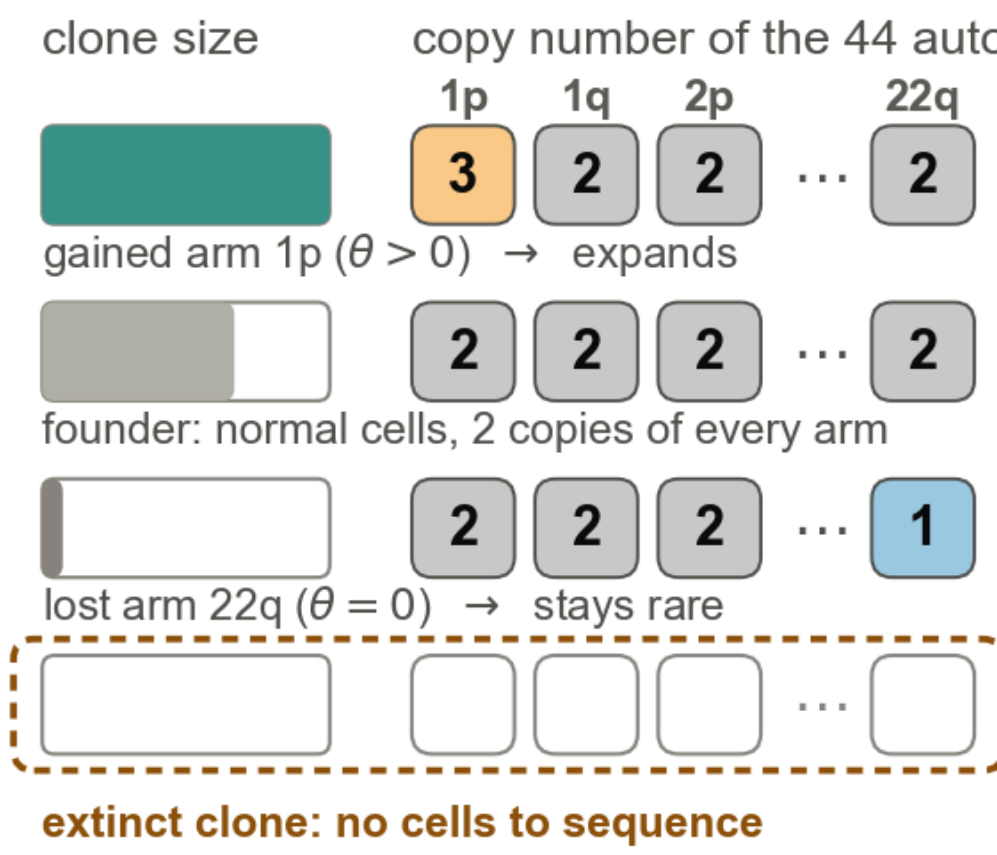
What actually happened



single-cell sequencing

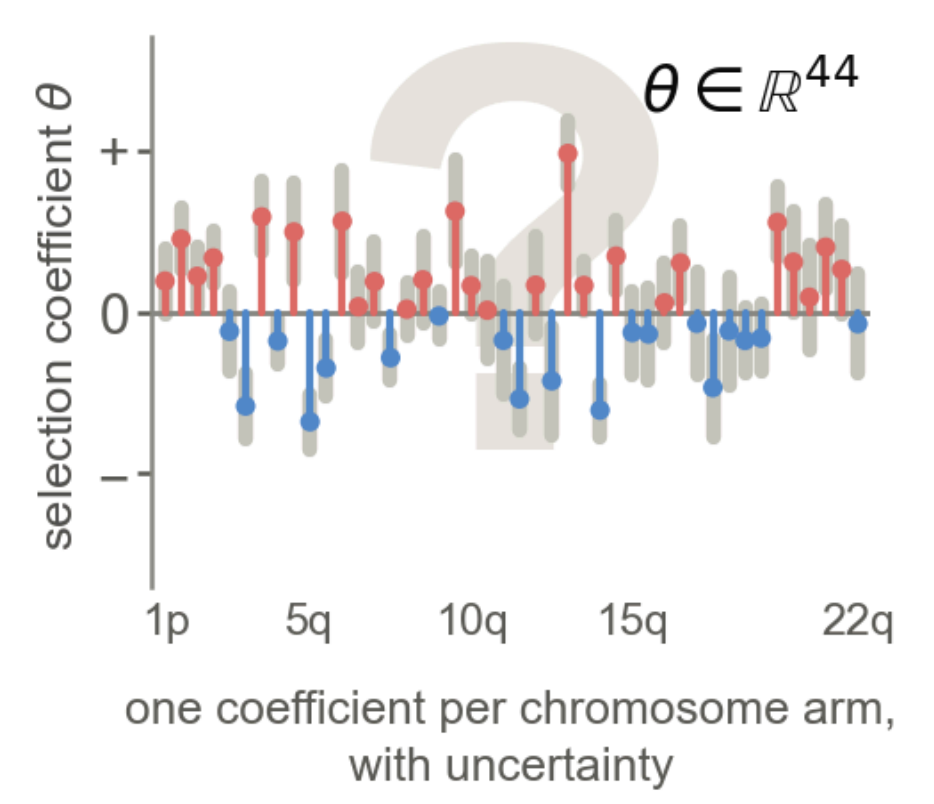
What we can observe

one time point · clone sizes f + the copy number X of each arm



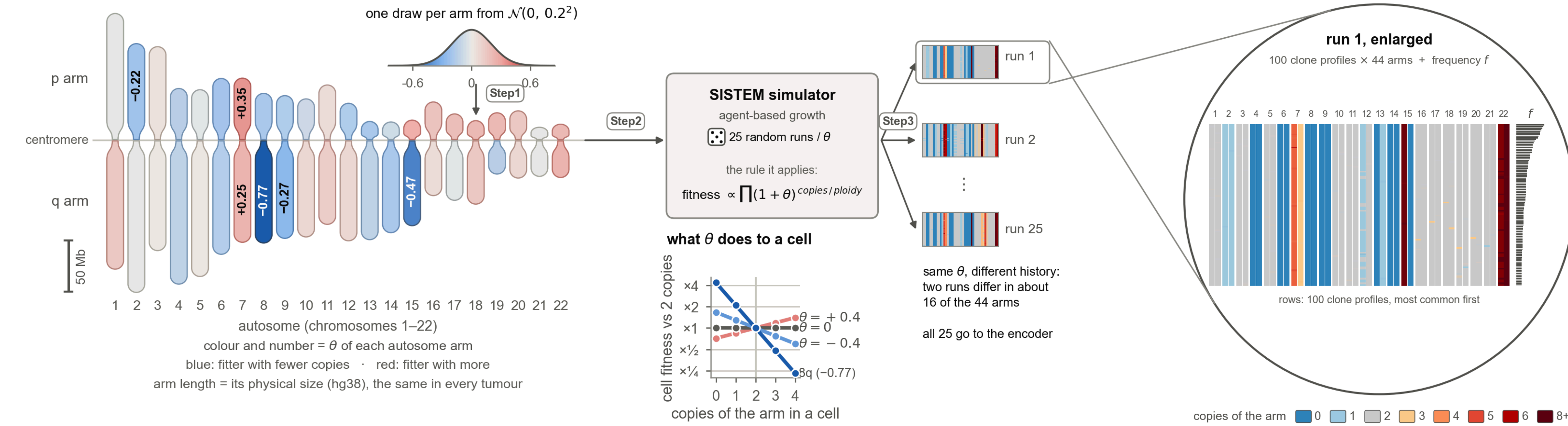
inverse problem

What we want to know



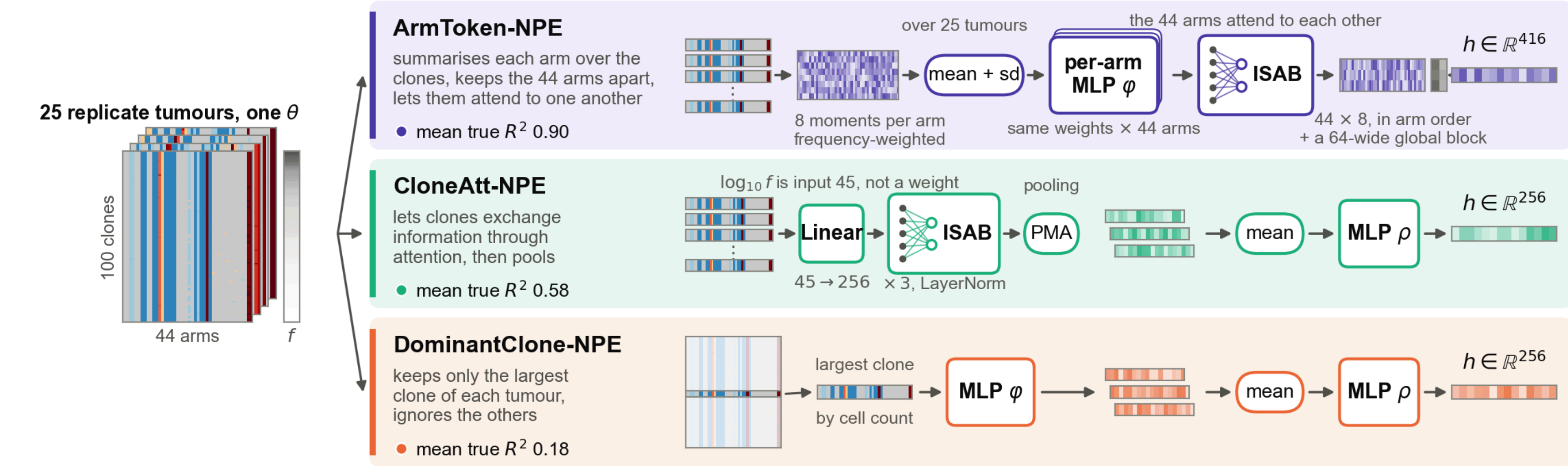
DATA: SIMULATED TUMOURS

Step1: draw one θ per autosome arm $\rightarrow$ **Step2:** SISTEM grows tumours under it $\rightarrow$ **Step3:** 25 replicate tumours; the 100 most common clone profiles of each tumour and their frequencies f are what the model sees



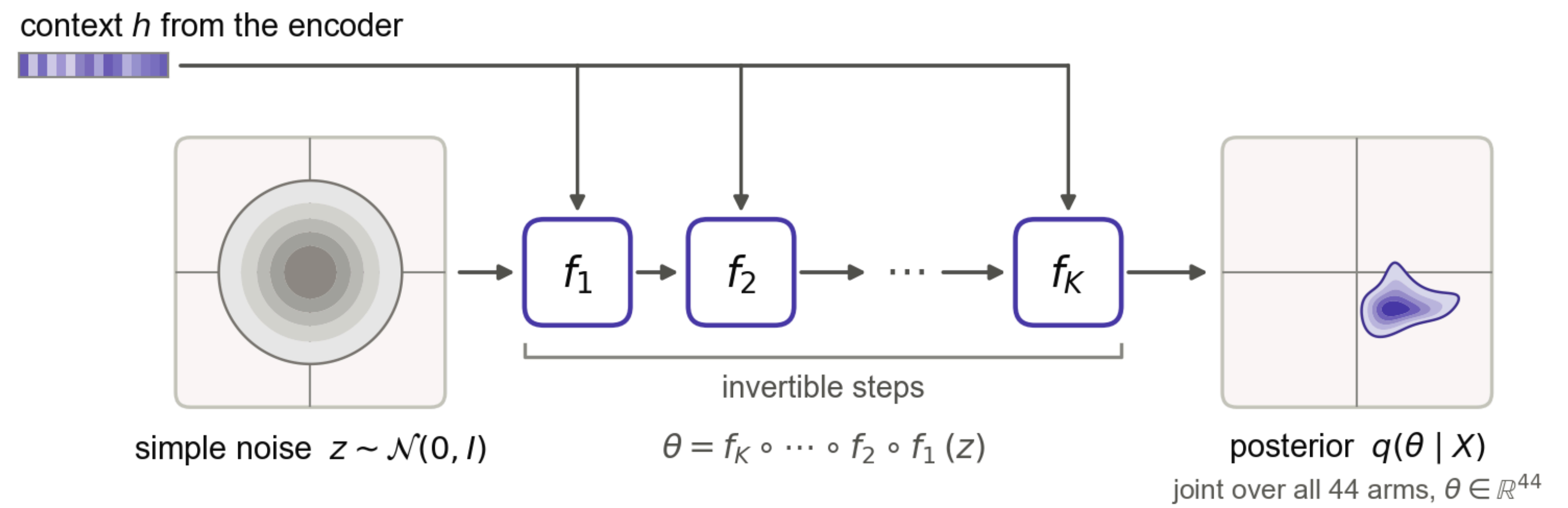
METHODS: ENCODERS

three ways to compress the 25 tumours into one context vector h : **CloneMLP** (every clone, frequency-weighted mean), **CloneAtt** (Set Transformer), **DominantClone** (largest clone only)



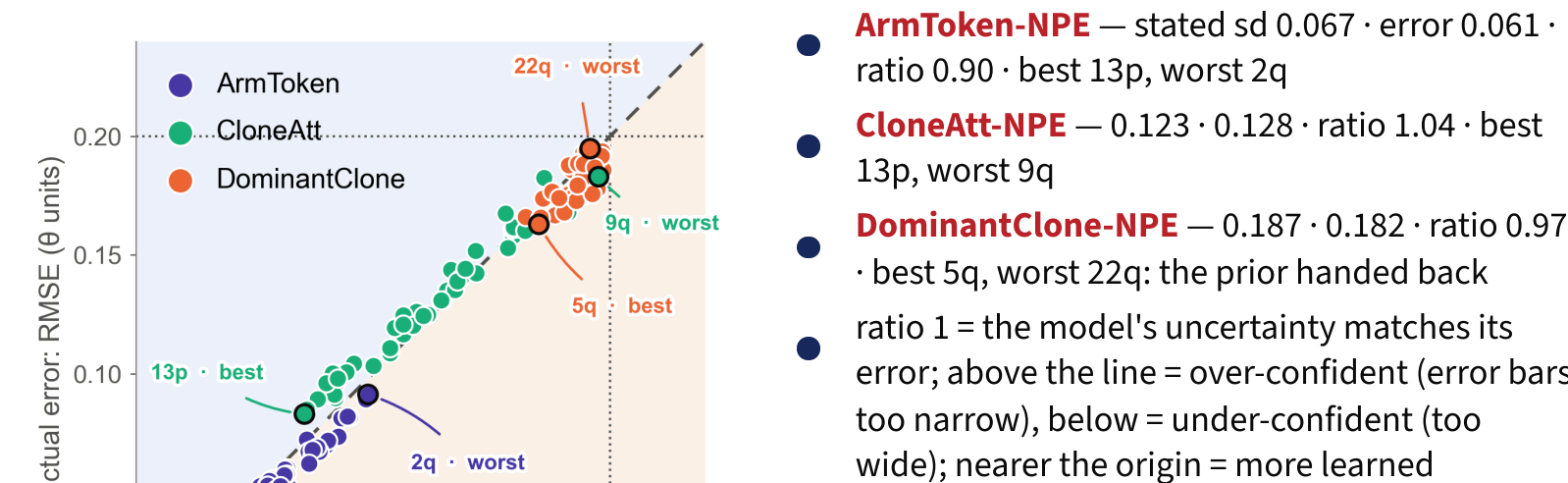
METHODS: POSTERIOR

a neural spline flow ($K = 5$ invertible steps, sbi) turns noise into $q(\theta | X)$, steered by h · trained on simulations only · amortised: instant for a new tumour

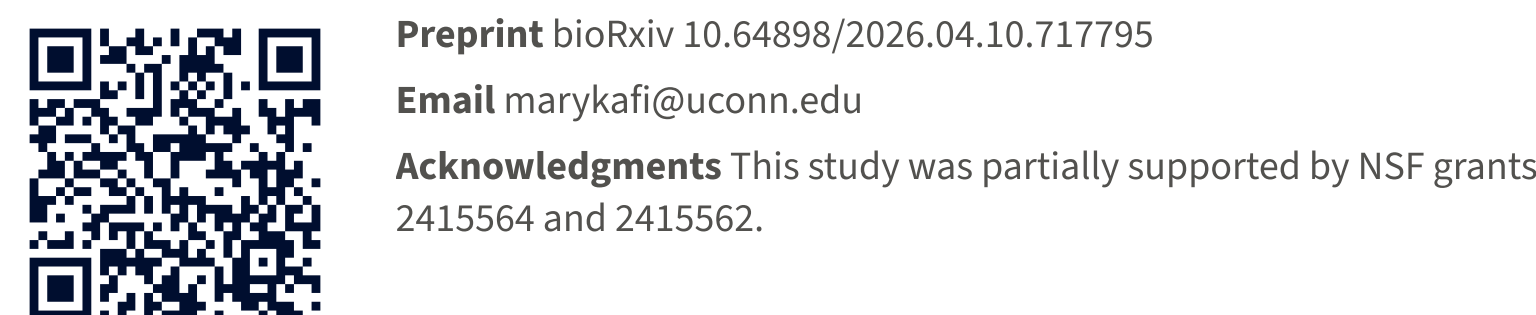


RESULTS: UNCERTAINTY VS ERROR

one dot per arm, both axes in θ units · dashed = honest (stated sd equals real error) · dotted = prior sd 0.20



MORE INFORMATION

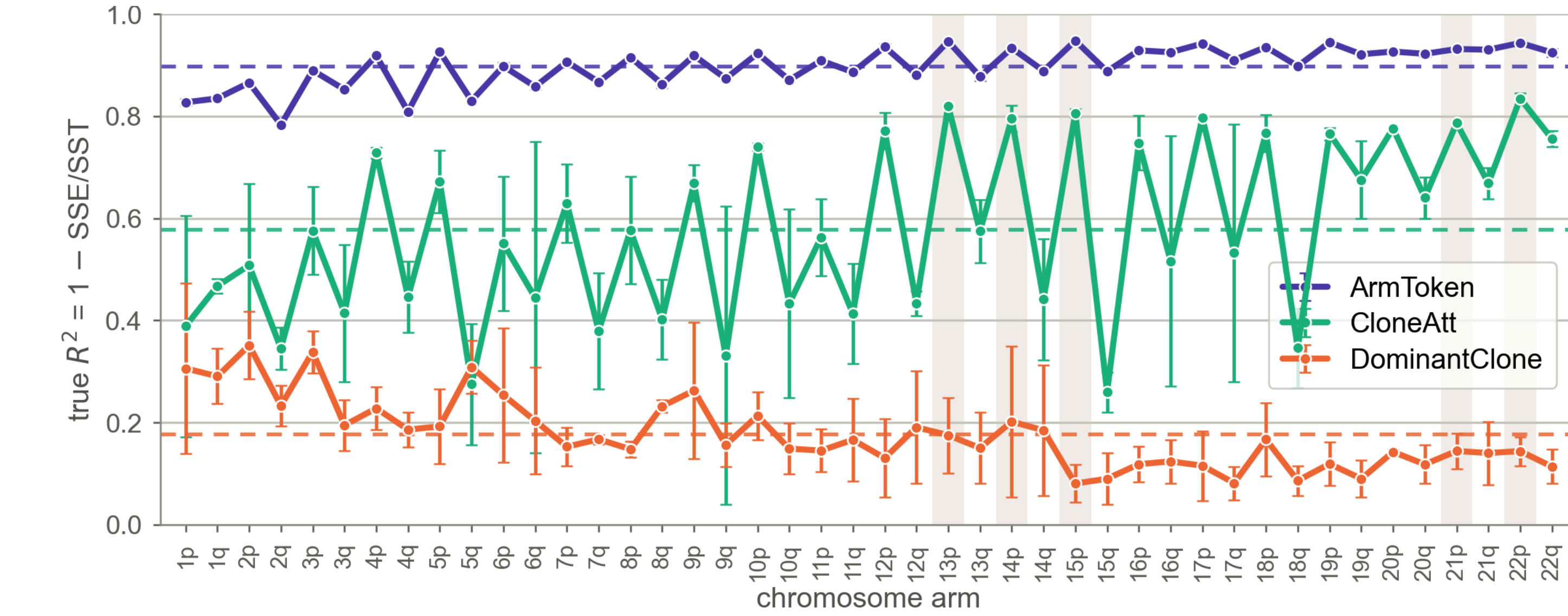


DISCUSSION

- **Feasible:** arm-level selection can be read from a single clonal snapshot, with uncertainty, without a likelihood.
- **Caveats:** the model was trained and tested with 25 replicate tumours per θ — a patient gives one, so it must be evaluated with a single tumour; the inputs are exact simulator copy numbers, whereas scDNA-seq samples cells and calls copy numbers with noise — simulating that step is the next thing to add.
- **Next:** test the model on real tumours.

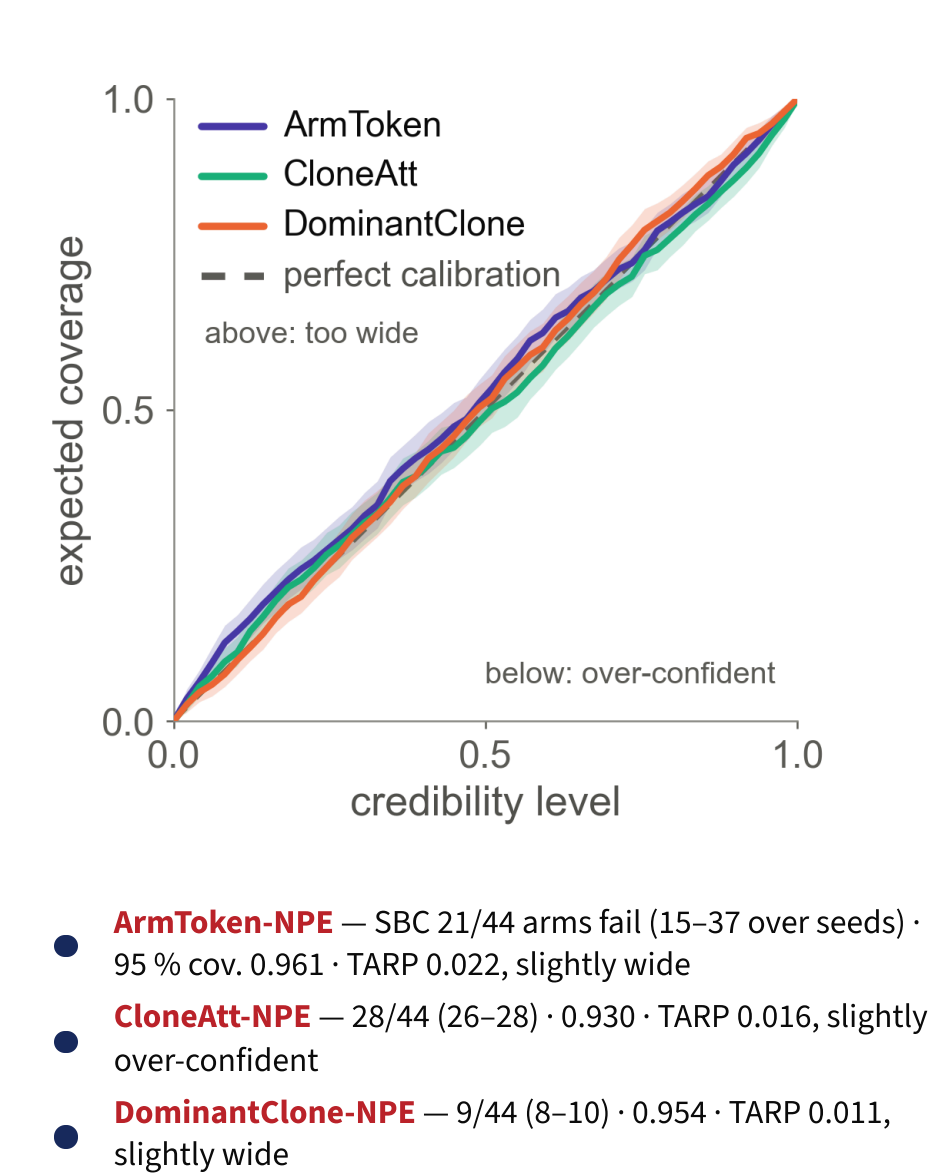
RESULTS: RECOVERY PER ARM

per-arm true R^2 ($1 - \text{SSE}/\text{SST}$) on 651 held-out simulated tumours · dashed = each encoder's mean over the 44 arms: **ArmToken-NPE 0.90** (3 seeds), **CloneAtt-NPE 0.58** (2), **DominantClone-NPE 0.18** (3) · error bars = sd over training seeds · shaded = acrocentric p arms



RESULTS: CALIBRATION

joint coverage over all 44 arms (TARP) · diagonal = honest



RESULTS

- **The clone set carries the signal:** the largest clone alone recovers θ with true R^2 0.18; attention over all clones 0.58.
- **Sharing weights across arms unlocks it:** ArmToken-NPE reaches 0.90 (± 0.002 over 3 seeds), ahead on 44 of 44 arms; the published model scored 0.47 with over-confident intervals.
- **Honest uncertainty:** all three encoders sit on the TARP diagonal; ArmToken's 95 % intervals cover 96 % and are 4x narrower than the prior.
- **Arms differ:** p arms beat q arms, and accuracy rises with chromosome number — but falls for the largest-clone baseline.
- 651 held-out simulated tumours; seed noise ± 0.002 (ArmToken), ± 0.015 (CloneAtt), ± 0.017 (DominantClone).

REFERENCES

- Salehi S., Kabeer F., Ceglia N. et al. Clonal fitness inferred from time-series modelling of single-cell cancer genomes. *Nature* 595, 585–590 (2021).
- Weiner S., Bansal M. S. SISTEM: simulation of tumor evolution, metastasis, and DNA-seq data under genotype-driven selection. *Bioinformatics* 41(12), btaf634 (2025).
- Greenberg D., Nonnenmacher M., Macke J. Automatic posterior transformation for likelihood-free inference. *Proc. ICML*, PMLR 97, 2404–2414 (2019).
- Durkan C., Bekasov A., Murray I., Papamakarios G. Neural spline flows. *Advances in Neural Information Processing Systems* 32, 7511–7522 (2019).
- Tejero-Cantero A., Boelts J., Deistler M. et al. sbi: a toolkit for simulation-based inference. *J. Open Source Softw.* 5(52), 2505 (2020).
- Lee J., Lee Y., Kim J., Kosiorek A., Choi S., Teh Y. W. Set Transformer: a framework for attention-based permutation-invariant neural networks. *Proc. ICML*, PMLR 97, 3744–3753 (2019).
- Boelts J., Deistler M., Gloeckler M. et al. sbi reloaded: a toolkit for simulation-based inference workflows. *J. Open Source Softw.* 10(108), 7754 (2025).